

JIANMAN LIN

Embodied AI · Vision-Language-Action Models · Generative Modeling

linjianmancjx@gmail.com | [Homepage](#) | [Google Scholar](#)

RESEARCH PROFILE

Master's student at SCUT, advised by [Prof. Tianshui Chen](#) and [Prof. Chunmei Qing](#). My work spans multimodal representation learning and controllable diffusion generation, and now centers on **Vision-Language-Action (VLA) models for robotic manipulation**: full-stack real-robot deployment paired with research on why learned action generation fails out of distribution.

- **Publications**: 6 papers; first or co-first author on 4 — **TPAMI, IJCV, TIP, CVPR (Highlight, top 2.8%)**.
- **Real-Robot Systems**: Deployed a VLA policy that autonomously completes a ~5-minute long-horizon manipulation task.
- **Current Research**: Decoupling *how to move* from *where to move* — a goal-conditioned action prior for OOD-robust VLA.
- **Service & IP**: Reviewer for CVPR, NeurIPS, ECCV; 3 Chinese patents (2 granted).

EDUCATION

South China University of Technology (SCUT)

M.Eng. in Electronic Information · GPA: 80/100

Guangzhou, China

Sep. 2024 – Present

Guangdong University of Technology (GDUT)

B.Eng. in Industrial Engineering · GPA: 87.2/100

Guangzhou, China

Sep. 2020 – June 2024

SELECTED PUBLICATIONS

*Equal contribution. Author name in **bold**. Each title links to the paper or project page.

- [1] [Learning Spatial-Temporal Coherent Correlations for Speech-Preserving Facial Expression Manipulation](#)
Tianshui Chen, **Jianman Lin***, Zhijing Yang, Chunmei Qing, Liang Lin.
IEEE TPAMI, 2026.
- [2] [Neural Scene Designer: Self-Styled Semantic Image Manipulation](#)
Jianman Lin, Tianshui Chen, Chunmei Qing, Zhijing Yang, Shuangping Huang, Yuheng Ren, Liang Lin.
IEEE TIP, 2025.
- [3] [Contrastive Decoupled Representation Learning and Regularization for Speech-Preserving Facial Expression Manipulation](#)
Tianshui Chen, **Jianman Lin***, Zhijing Yang, Chunmei Qing, Yukai Shi, Liang Lin.
IJCV, 2025.
- [4] [Learning Adaptive Spatial Coherent Correlations for Speech-Preserving Facial Expression Manipulation](#)
Tianshui Chen, **Jianman Lin**, Zhijing Yang, Chunmei Qing, Liang Lin.
CVPR, 2024. **Highlight (top 2.8%)**
- [5] [Personalized Cross-Modal Emotional Correlation Learning for Speech-Preserving Facial Expression Manipulation](#)
Tianshui Chen, Ying Zhu, **Jianman Lin**, Zhijing Yang, Chunmei Qing, Fen Gao, Liang Lin.
IEEE TMM, 2026.
- [6] [Geometry-Editable and Appearance-Preserving Object Composition](#)
Jianman Lin, Haojie Li, Chunmei Qing, Zhijing Yang, Liang Lin, Tianshui Chen.
arXiv preprint arXiv:2505.20914, 2025.

RESEARCH EXPERIENCE

Current Research: Goal-Conditioned Action Prior for OOD-Robust VLA

2026 – Present

Question: Why do VLA policies fail out of distribution, and can motion generation be made reusable across scenes?

- **Finding**. Attention analysis on a PI05-style VLA (pretrained VLM + a from-scratch flow-matching Action Expert) shows that zero-shot OOD failures are *not* perception failures: the VLM still localizes the target and the Action Expert still attends to it. The failure appears when that information is turned into an executable trajectory.

- **Insight.** Visual conditioning from the very start of training makes visual features govern not only *where* to move but also *how* the trajectory is generated, coupling action generation to the training visual distribution.
- **Direction.** I am developing a formulation that decouples motion generation from visual conditioning, so that vision specifies the spatial goal while a reusable action prior supplies the motion.

Speech-Preserving Facial Expression Manipulation [1, 3, 4]

2023 – 2025

Focus: Addressing data scarcity by mining intrinsic priors and learning decoupled representations.

- Learned adaptive spatial-temporal coherent correlations between local visual disparities to supply supervision under limited paired data, then extended the framework to video for temporally robust manipulation.
- Designed contrastive decoupled representation learning (CCRL/CERL) to disentangle content from emotion features.

Controllable Image Editing via Diffusion-Based Learning [2, 6]

2024 – 2025

Focus: Resolving the trade-off between editability and visual consistency in diffusion-based editing.

- *Neural Scene Designer*: a diffusion framework combining dual parallel cross-attention with self-style representation learning for style-consistent semantic manipulation.
- *DGAD*: decouples geometric modeling from appearance alignment via dense cross-attention retrieval for object composition.

WORK EXPERIENCE

Research Intern: Embodied AI & Robotics

Dec. 2023 – Present

X-era.ai & ExpanderaAI Lab

Guangzhou, China

2026 – Present

- **Embodied AI and Robotics**
 - Trained and deployed a VLA policy achieving stable autonomous execution of a ~5-min long-horizon task.
 - Core algorithm researcher for long-horizon, fine-grained manipulation, owning the full stack: hardware setup, teleoperation data collection, training, real-world deployment, and iterative optimization.
 - Hands-on with AgileX Piper, UR5, ARX and Zerith arms: teleoperation, control code, cross-platform integration.
- **Intelligent Furniture Project** 2024 – 2025
 - Core algorithm researcher for the Feiliu AI Smart Interior Design system [[Product Link](#)]: VLM-based floorplan parsing, LLM-based layout generation, and an embodied design agent.
- **Digital Human Live Streaming** 2023 – 2024
 - Core algorithm member across the full pipeline: data processing, model training, and online deployment.
 - Used high-resolution StyleGAN facial priors for few-shot adaptation to identity-consistent avatars.

PATENTS

- [1] Method for Image Texture Transfer Based on Temporal Decoupled Diffusion Model
Jianman Lin, Chunmei Qing, Tianshui Chen. *CN121685688A*, 2026 (under substantive examination).
- [2] Method and System for Webly-Supervised Fine-Grained Image Recognition Based on Deep Learning
Jianman Lin, Tianshui Chen, Jiantao Lin, Zhijing Yang. *CN115496948B*, 2025 (**Granted**).
- [3] Method, Apparatus, Device, and Storage Medium for Stylized Image Generation
Ke Gong, **Jianman Lin**. *CN118014820B*, 2024 (**Granted**).

ACADEMIC SERVICE & TECHNICAL SKILLS

Reviewer: CVPR, NeurIPS, ECCV.

Embodied AI & Robotics: VLA models, robotic manipulation, multi-arm pipelines, teleoperation, real-world closed-loop deployment.

Deep Learning & Generative AI: PyTorch, HuggingFace (Transformers, Diffusers), DeepSpeed, TensorRT/ONNX.

Programming & Infrastructure: Python, C/C++, CUDA, Linux, Git, Docker.